Low Code Engineer (OutSystems Platform)

GovTech is the lead agency driving Singapore's Smart Nation initiatives and public sector digital transformation. As the Centre of Excellence for Infocomm Technology and Smart Systems (ICT & SS), GovTech develops the Singapore Government's capabilities in Data Science & Artificial Intelligence, Application Development, Smart City Technology, Digital Infrastructure, and Cybersecurity.

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About Ministry of Social and Family Development (MSF)

The Ministry of Social and Family Development (MSF) is Singapore's lead ministry for social and family policies, working to build a caring and inclusive society where every individual and family can thrive.

We deliver essential services through our network of Family Service Centres, Social Service Offices, and industry partners, supporting vulnerable populations across all life stages. Our work encompasses family services, early childhood, social assistance, disability support, mental health initiatives, and community development programmes.

Position Overview

We are seeking a talented Low Code Engineer who combines deep OutSystems expertise with strong full-stack engineering fundamentals to deliver and operate secure, reliable, citizen-facing digital services.

You will work across application architecture, integrations, SQL Server, IIS, AWS/GCC infrastructure, security, observability and CI/CD; troubleshoot complex production issues; and drive modernisation, automation and engineering standards across the platform.

Key Responsibilities

Application Engineering

- Design, build and maintain secure, scalable applications and integrations using APIs and SFTP.
- Translate business needs into maintainable technical solutions and improve existing applications for quality and performance.
- Contribute to architecture, technical design and code reviews.

Engineering Excellence

- Advance CI/CD, automated testing, observability and operational automation.
  Promote secure, maintainable designs, engineering standards, documentation, knowledge sharing and platform governance.

Platform Operations & Reliability

- Operate OutSystems environments across development, testing and production, including deployments, upgrades, migrations and environment management.

- Diagnose and resolve application, workflow, timer, integration and platform issues using logs, metrics and monitoring data.
- Support production incidents, perform root cause analysis and implement preventive improvements.

Cloud, Infrastructure & Security

- Design, build and maintain services hosted on AWS/GCC and troubleshoot IIS, Windows Server, SQL Server, networking and related infrastructure.
- Contribute to compliance, hardening, backup, patching and operational resilience activities.

Key Requirements

OutSystems Development

- At least 4 years of experience developing and maintaining applications using the OutSystems platform.
- Strong understanding of OutSystems architecture, LifeTime, Service Center, deployment processes and development best practices.
- Experience designing secure, scalable and maintainable OutSystems applications and integrations.
- Ability to troubleshoot application, workflow, timer, integration and platform issues in production environments.

Full-Stack Engineering

- At least 6 years of professional software engineering experience, with strong fundamentals in modern application architecture and design.
- Working knowledge of C# and .NET, with experience designing and developing backend services and APIs.
- Proficiency in modern frontend development using JavaScript or TypeScript, HTML5, CSS3, and frameworks such as React or Vue.
- Strong experience with Microsoft SQL Server, including data modelling, complex queries, performance optimisation and production troubleshooting.
- Experience implementing integrations using REST, SOAP or other API technologies.
- Experience supporting applications on AWS, GCC or equivalent cloud platforms, with working knowledge of IIS, Windows Server, networking, security, high availability and disaster recovery.
- Experience with Git, CI/CD, automated testing, monitoring, logging, observability and production incident management; familiarity with Infrastructure as Code is an advantage.
- Understanding of secure software development, including authentication, authorisation, encryption, audit logging and compliance requirements in regulated environments.
- Ability to troubleshoot across application, database, integration and infrastructure layers and drive issues through to resolution.

Key Attributes

- **Engineering Ownership:** Demonstrate experience and track record of accountability for outcomes and driving issues & opportunities, through to resolution.
- **Engineering Excellence:** Produces maintainable, reliable, and secure solutions while continuously improving engineering practices.
- **Deep Problem Solving:** Diagnoses issues across application, database, integration and infrastructure layers, perform root-cause analysis well and independently.
- **Trade-offs & prioritisation:** Balances engineering quality, business priorities and operational constraints.

- **Collaboration & Communication:** Communicates clearly and works effectively with product managers, engineers and designers. Shares trade-offs between speed, quality, and simplicity, communicating respectfully.
- **Continuous learning:** Demonstrates curiosity and adapts to latest technologies, engineering practices and industry trends. Implementation and/or integration with AI expected in the role

Application process and assessment

To apply for this role, candidates must submit the completed technical assessment together with their application. Candidates are encouraged to complete the assessment within three calendar days.

Objective

Design and develop a full-stack web application that consumes publicly available data from data.gov.sg using OutSystems.

High-Level Requirements

- Retrieve and integrate data from any dataset of your choice available on data.gov.sg.
- Present the data through a clear and user-friendly interface.
- Provide appropriate capabilities for users to explore and interact with the data.
- Demonstrate sound frontend and backend application design.
- Provide Infrastructure as Code (IaC) that defines the AWS infrastructure required to deploy and operate the OutSystems Platforms.
- Include appropriate automated testing.

Assessment Criteria

The assessment will focus on the overall quality of the solution, engineering practices, technical decision-making, and the candidate's ability to explain and justify their design choices, assumptions, and trade-offs.

Submission

1. Provide the access to the OutSystems Environment
2. Submit the complete Infrastructure as Code, automated tests and README documentation in a Git repository as part of your application. Provide the access to the repository.